



Ain Shams University
Faculty of Engineering
Computer Engineering and Software Systems Program
CSE432: Automata and Computability – Spring 2026

T E R M - P R O J E C T R E Q U I R E M E N T S

This project is a group project with each group has 4 to 6 students. A detailed description that elaborates on the proposed project is needed. The project should be in the following categories:

Part 1

Prepare a comprehensive report explaining how Turing Machines and Pushdown Automata are utilized within computational models. The report should include the following components:

- A description of their practical implementations
- An evaluation of their advantages and disadvantages
- A discussion of the types of problems they address or overcome
- An analysis of their limitations and drawbacks
- Any additional relevant insights or supporting information that enhance understanding of these models

Part 2

Write a java – C++ or C# program with suitable GUI to implement the following

1. convert context free grammar to Pushdown automaton
2. Construct a DFA over $\{0,1\}$ that accepts strings where the number of 1s is divisible by 3 **and** ends with 0.
3. Construct PDA for: $L = \{ a^n b^n \mid n \geq 0 \}$

In addition, provide detailed documentation for the developed programs, including screenshots that clearly demonstrate the program outputs and functionality.

Important Notes:

- Project source code must be submitted on LMS in a zip/rar archive. Full implementation of the project should be delivered by the due time.
- Use consistent document format (font sizes, titles, subtitles, captions, paragraph formatting ... etc.). Recommended font sizes are: main title 14pt, subtitles 12pt, main text 12pt, and captions 10 pt. Recommended font type is “**Arial Black**” for titles and subtitles, and “Arial” for all other texts. Recommended spaces before and after paragraphs are 12pt before and 6pt after each paragraph, and 1.5 spacing is highly recommended. Justified paragraphs from both sides are also recommended.
- Figures and tables must be centred in the pages, and they should be numbered separately. Each figure/table must have a caption that appears below the figure/table.
- Pages must be numbered consistently except the cover page.
- All reports must be written in English, always avoid typos and grammatical errors.
- Report will go plagiarism check.
- All project deliverable must be uploaded to the LMS, no hardcopy is accepted, and please do not send your project deliverables by email.